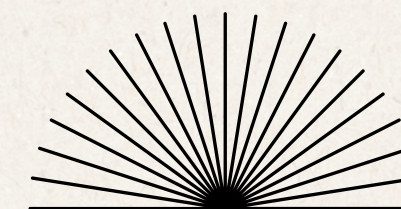




Indian Institute of Technology Kanpur

SIGNAL SPRINT HACKATHON

LeafyGreen - S Kiran



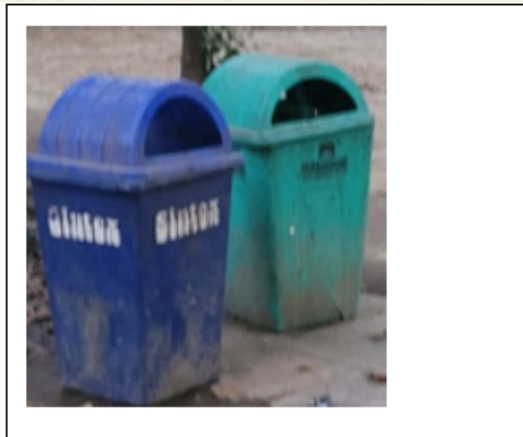
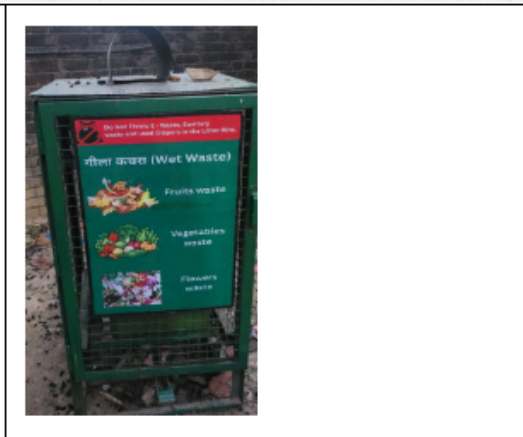
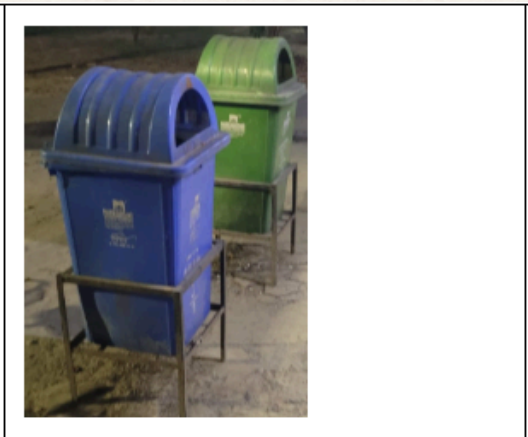
Problem Statement and Approach:

REQUIREMENTS

- to intimate DMC when a dustbin is in a spill or overflow condition. Or garbage is lying just near the dustbin but not inside it.
- Garbage is defined as non-useful and thrown manmade material objects.
- dustbin-centric
- model must understand context, not just detect garbage or dustbins

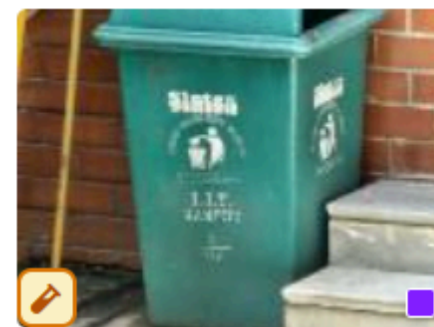
SOLUTION

- Build a CNN based image multi label classification: Overflow and Valid. Inform DMC when both positive
- Curate the Dataset such that random multi-coloured variations closely match with litter.
- stricter weightage on dustbin condition
- Implement a patch-wise learning to the model which can understand local context

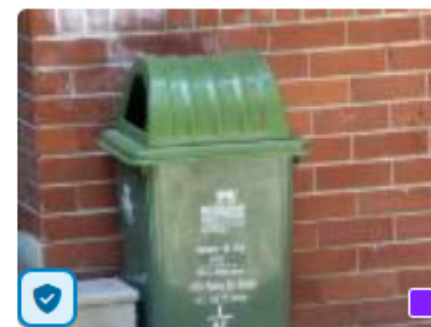
		
Blue/Green/Cyan and Standing on ground. Top lead may ON or OFF	Metal Cage fully packed dustbin	Metal stand elevated Blue/Green/Cyan dustbins

Dataset: Creation and Augmentation

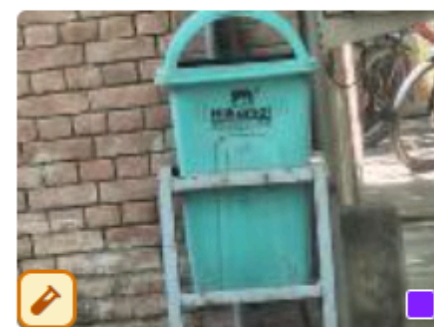
- Contains Mainly of 2 Multi-Label Classes Namely “Overflow” and “Valid”
- Obtained ~180 Images from in and around the campus consisting of valid & invalid bins, overflowing litter etc and ~300 images Handpicked to maintain similarity mentioned in the problem statement
- contains lots ample amount of samples with lighting variations- Day and Night, jittered and blurred images for robustness multiple angles for accurate learning



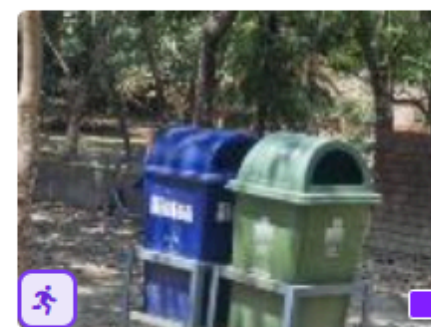
20260403_123244.jpg



20260403_123249.jpg



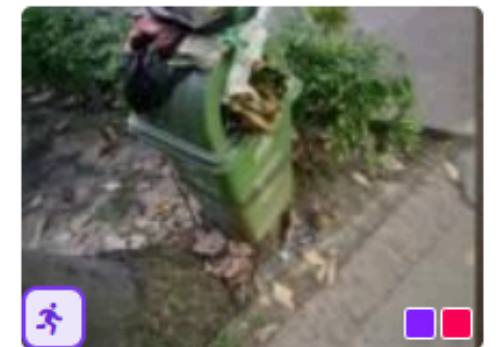
20260403_124005.jpg



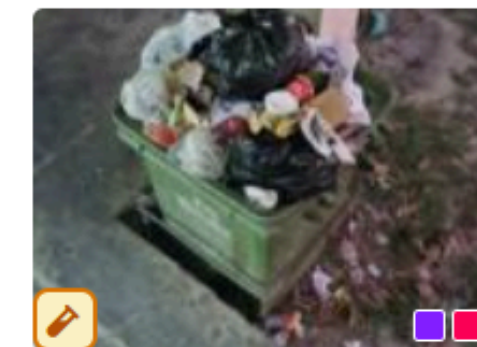
20260403_124233.jpg



1776090743002(4).png



1776090743002(5).png



1776091388931(1).png



1776091388931(2).png

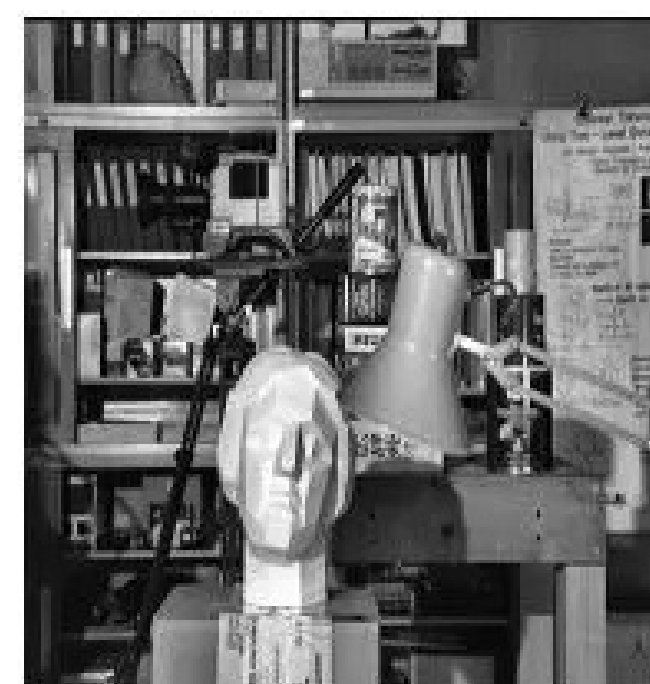
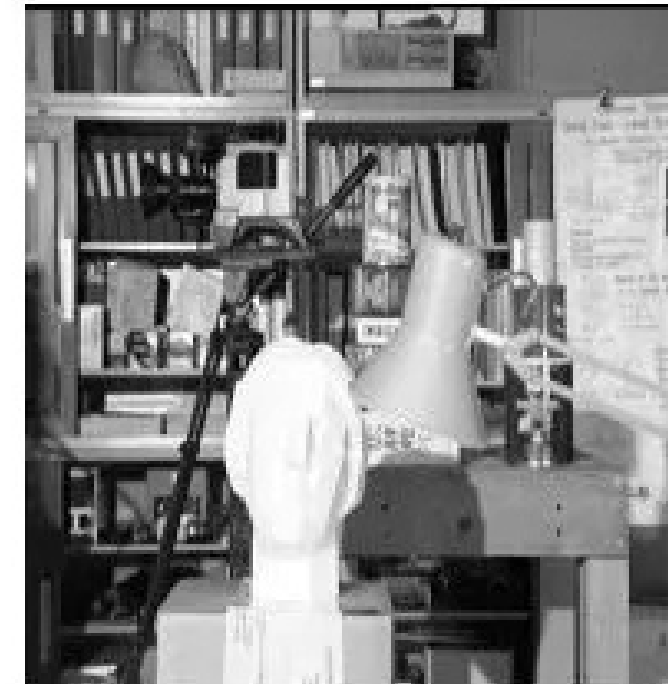
Dataset: Creation and Augmentation

- Augmented net image dataset to obtain a larger sample of limited image dataset for better accuracy of the model.
- Applied image stretch to 512x512, Horizontal Flips, Rotation Between -11° and $+11^\circ$, Shear: $\pm 13^\circ$ Horizontal, $\pm 12^\circ$ Vertical, Grayscale: Apply to 18% of images, Brightness: Between -15% and +15%.
- Used random aggregation of images of same class to overcome edge and challenging cases



Dataset: Image Split and Pre-Process

- Apply an adaptive histogram equalization technique to bring out sharper edges and contrast for better inference
- Use standard 70:20:10 split for accurate analysis of model
- if dataset skew is higher use a curated focal_loss for accuracy metric instead of standard Binary_Crossentropy



```
Train: 1293  Val: 123  Test: 61
```

```
Train label distribution:  
valid  overflow
```

```
0      0      354
```

```
1      0      327
```

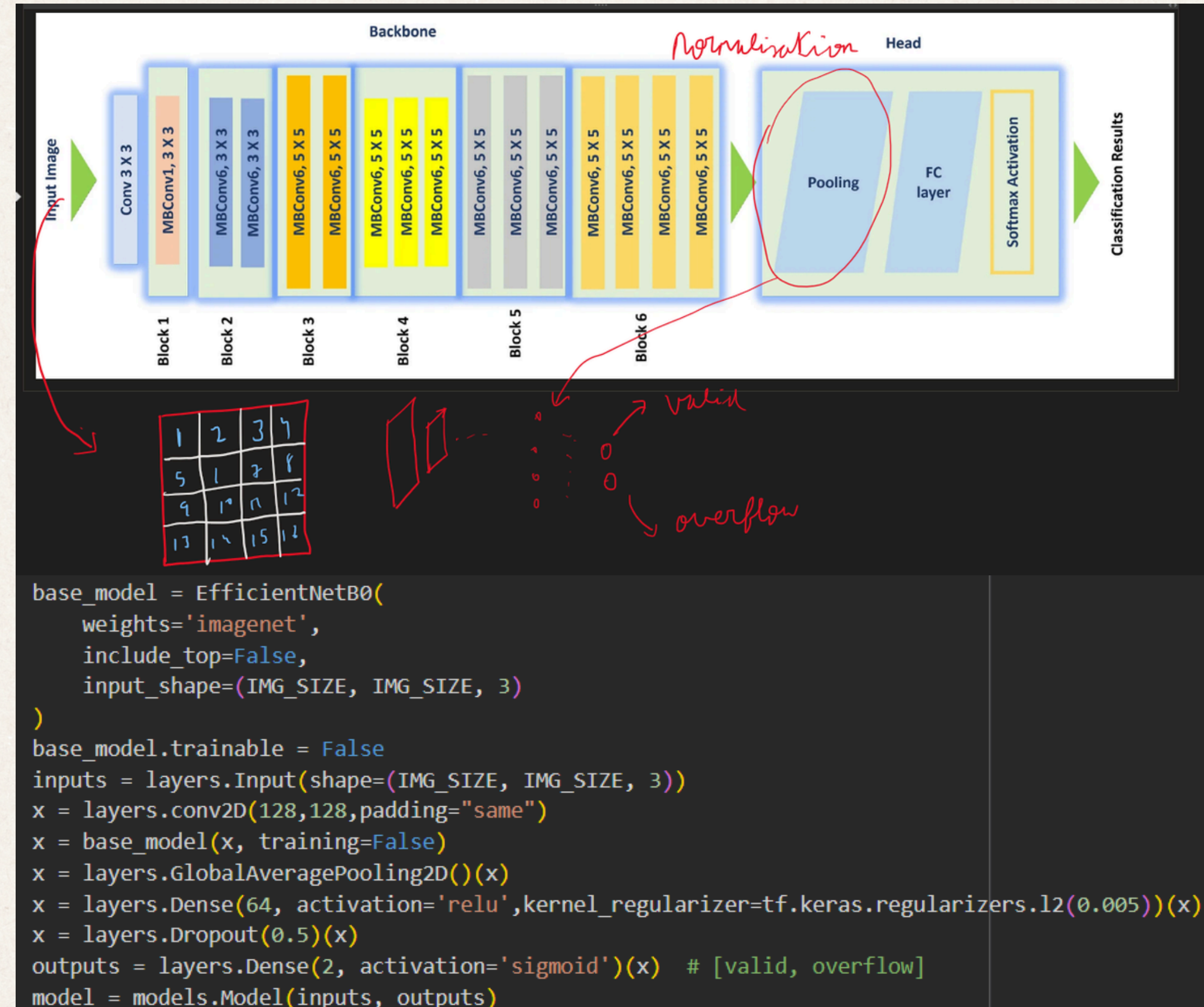
```
0      1      312
```

```
1      1      300
```

```
Name: count, dtype: int64
```


Model: Architecture

Input image consists of 512x512 sized images → convolution layer into 16 patches -
> EfficientNetB0 backbone → global average pooling → ANN heads → 2 neuron sigmoid classification layer. for multiple inference learning an Attention Based model provided HEAVY augmentation is applied for better learning.



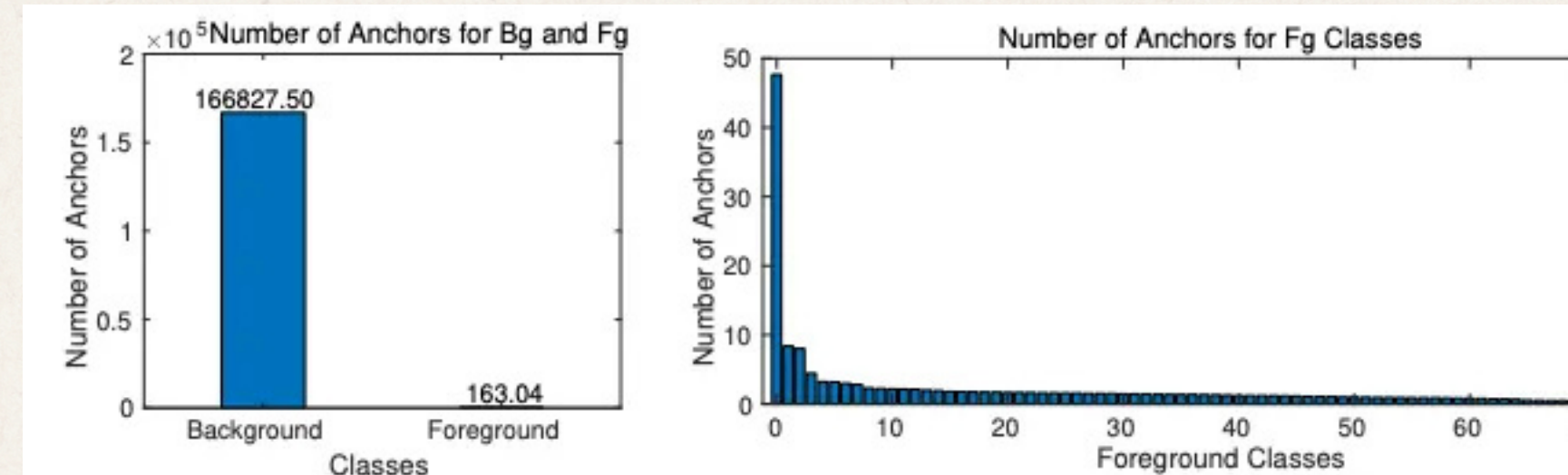
Model: Loss Function

Focal Loss:

- Mainly used when there exists high disparity between the classes involved, used for focused learning
- here p_t represents the predicted probability of the class, Alpha handle's the class imbalance problem, gamma indicates the order of loss (higher gamma $\rightarrow$ disparity towards harder cases from the dataset)

Weighted BCE:

- Equivalent to BCE but handles class weights parameters depending on the variety in dataset
- w_1, w_2 shows the weights added to the BCE of each (of the two) classes to focus more on class having a higher weight.

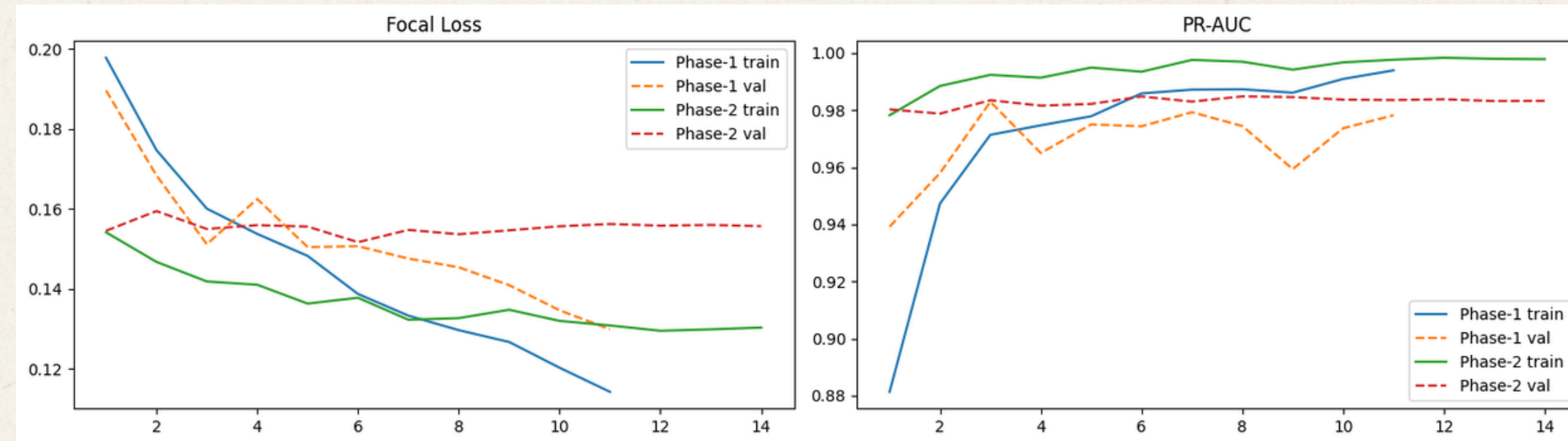


$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

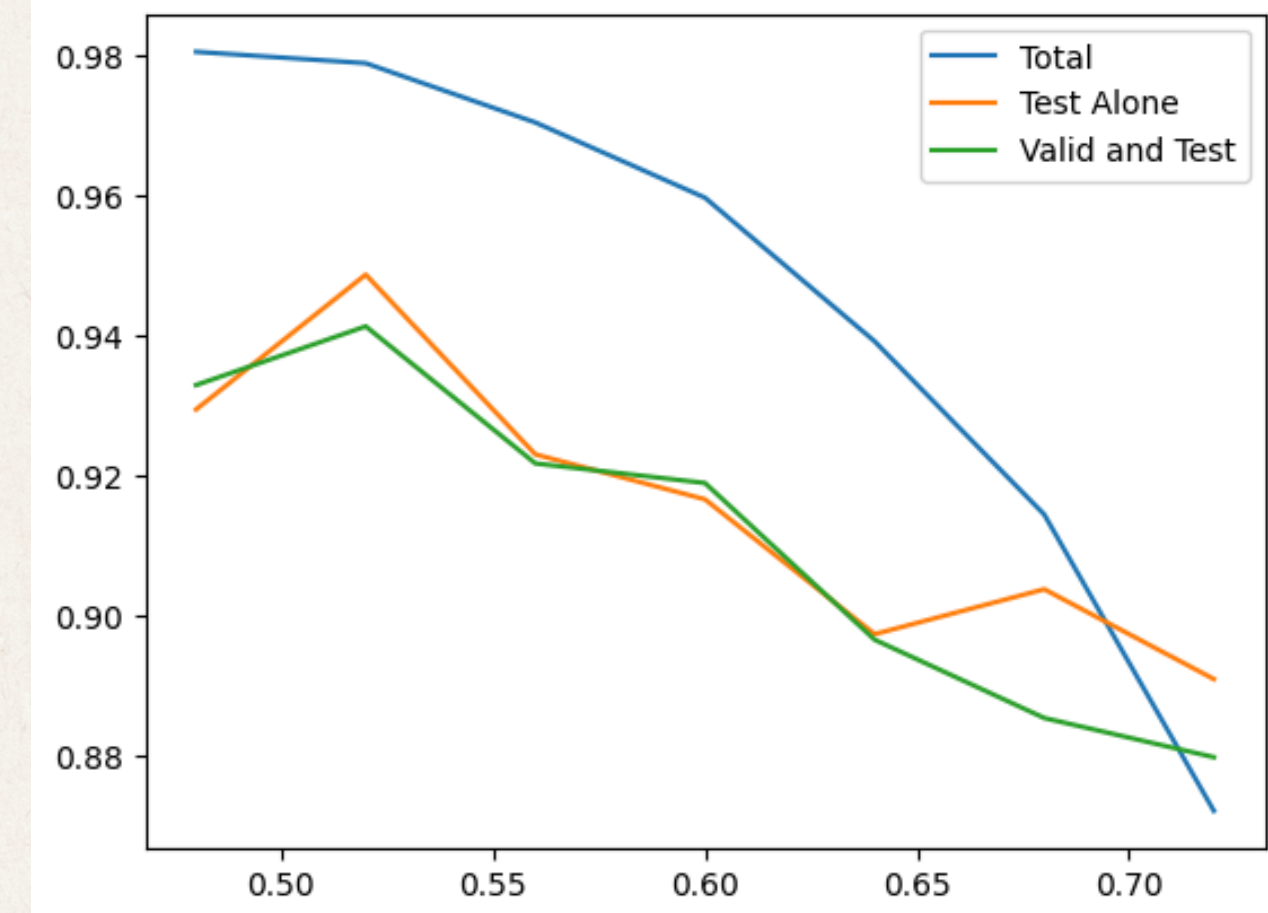
$$w_I \frac{1}{N} \sum_{i=1}^N y_i \cdot \log(p(y_i)) + (1 - y_i) \cdot \log(1 - p(y_i))$$

Model: Accuracy and Tuning:

Iterate the model through various threshold values to increase accuracy and F1 score to reduce false negatives hence misclassifying overflow as not.

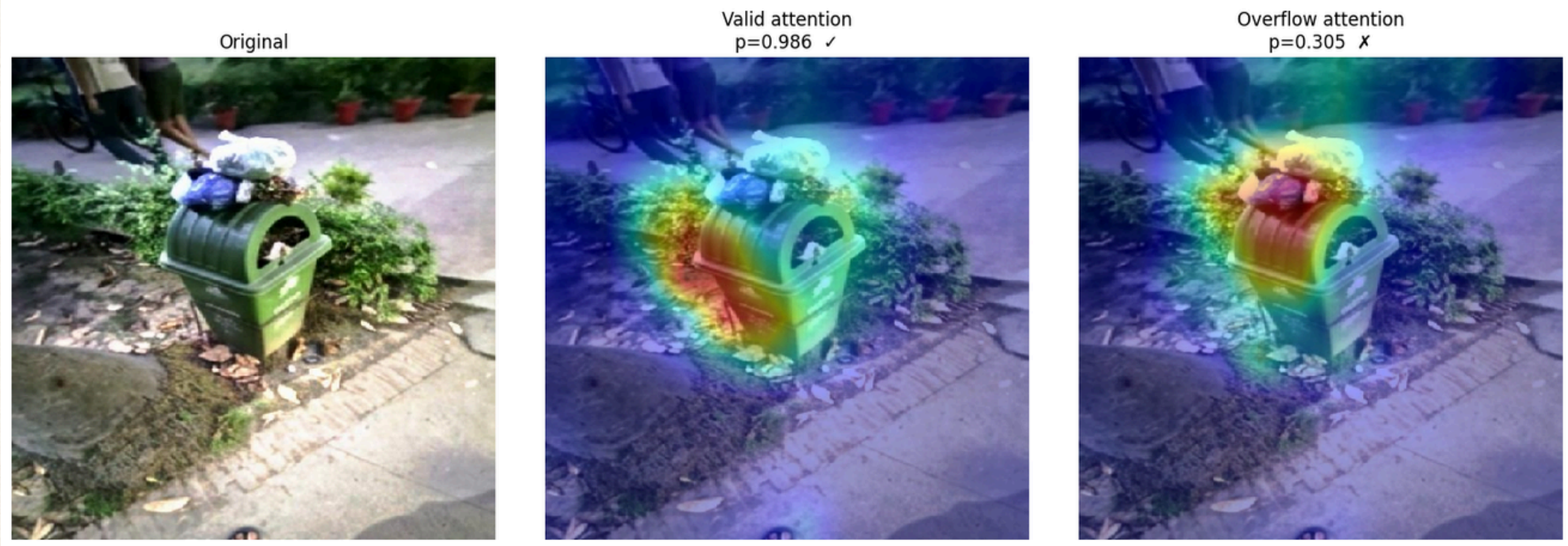


obtaining the maximum from the given graph lets us decide the apt threshold for higher accuracy using, weighted BCE mentioned previously



Model: Results & Final:

After the initial fine tuning we can see that the model now is able to verify between what is valid and what is overflow, with high accuracy and precision. With a more robust dataset(valid bins) the accuracy can be boosted further.



Final Score obtained on given Test Dataset(made)
and given public : 9.0/10

Test metrics:				
accuracy: 0.9262				
loss: 0.1508				
pr_auc: 0.9862				
precision: 0.9630				
recall: 0.8814				
Valid				
	precision	recall	f1-score	support
not valid	0.91	0.94	0.93	33
valid	0.93	0.89	0.91	28
accuracy			0.92	61
macro avg	0.92	0.92	0.92	61
weighted avg	0.92	0.92	0.92	61
Overflow				
	precision	recall	f1-score	support
not overflow	0.88	1.00	0.94	30
overflow	1.00	0.87	0.93	31
accuracy			0.93	61
macro avg	0.94	0.94	0.93	61
weighted avg	0.94	0.93	0.93	61

Website:

Visit the website, Simple UI, just select a model and upload or capture a picture of the affected bin, according to verdict on notification DMC would be called

Welcome User 🙌

IITK DMC Overfill Portal

To intimate DMC utilities, upload or click a picture of Overfilled Dustbin area:

File's Uploaded : True

Validity Percentage Confidence : 0 False

Overflow Percentage Confidence : 0 False

Final Verdict on Notification : None

Thank's for playing your part in keeping the campus clean! 🗑️



Choose ML model:

☒ Model_1.0 ☐ Model_2.0 ☐ Model_Tuned

Choose source:

☒ Upload Photo ☐ Take Photo

Upload Image

 Upload

200MB per file • JPG, JPEG, PNG

< Manage